Uncertainty intervals

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Uncertainty intervals before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Sampling variability

Without opening the old lesson, explain one key idea from Sampling variability. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Experimental units and independence

Without opening the old lesson, explain one key idea from Experimental units and independence. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

A literature comparison matrix

Without opening the old lesson, explain one key idea from A literature comparison matrix. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Uncertainty intervals trace and record the first step you can explain without help.

Use W2: Trace the worked case for Uncertainty intervals. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Uncertainty intervals explanation against the supplied facts before discussing it.

Use the worked case for Uncertainty intervals. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Observed mean is 20 from 25 independent normal measurements; population standard deviation is known to be 5. Use the supplied 95% rule mean ± 1.96 × sigma / sqrt(n). Compute the interval and name one assumption.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Uncertainty intervals facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Uncertainty intervals facts and keep your reasoning.

Assume 16 independent normal measurements with observed mean 10 and known population standard deviation 2. Use a 95% interval: mean ± 1.96 × sigma / sqrt(n). Compute the **standard error**, margin and interval. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

W3 Audit the limit

> Audit the limit. Apply the supplied Uncertainty intervals facts and keep your reasoning.

A report omits this qualification: A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Uncertainty intervals facts and keep your reasoning.

Observed mean is 20 from 25 independent normal measurements; population standard deviation is known to be 5. Use the supplied 95% rule mean ± 1.96 × sigma / sqrt(n). Compute the interval and name one assumption.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Uncertainty intervals facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Uncertainty intervals to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled uncertainty intervals **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of uncertainty intervals.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Uncertainty intervals

> Read a supplied uncertainty interval. A fictional study supplies an interval for a mean time difference under its stated sampling model. Explain the labelled step before checking the changed condition.

Read a supplied uncertainty interval. A fictional study supplies an interval for a mean time difference under its stated sampling model.

Estimated difference B–A is -0.8 seconds; supplied 95% **confidence interval** is [-1.4,-0.2] seconds. Assume the study's interval method is appropriate.

AI ACADEMY / AI-7/12

Supplied interval for B minus A time

	Lower endpoint (s)	Estimate (s)	Upper endpoint (s)
Original supplied interval	-1.4	-0.8	-0.2
Changed supplied interval	-1.4	not specified	0.3

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/12

Uncertainty intervals

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Locate the estimate

-0.8 lies between the interval endpoints.



2. Interpret the sign

Negative B-A means B took less time under this measurement convention.



3. Respect coverage meaning

95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

CHANGE ONE THING

Use a supplied interval [-1.4,0.3] instead.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2]. Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Read a supplied uncertainty interval. A fictional study supplies an interval for a mean time difference under its stated sampling model.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? 95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

1. Interpret the sign

2. Locate the estimate

3. Respect coverage meaning

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Respect coverage meaning 95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

An uncertainty interval conveys precision under assumptions. Its endpoints, units, comparison direction and method all matter; the interval is supplied here, not calculated from unavailable raw data.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Use a supplied interval [-1.4,0.3] instead. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. It includes zero and positive differences, so both no difference and slower B remain compatible with that interval.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval. The new upper endpoint crosses zero.

Repair a missing building block

Review Sampling variability (ai-7/11). Goal: Explain the **mechanism** and terminology of sampling variability. Short practice: Calculate the two four-case rates and their difference.

Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.. Return to this week and explain what you changed.

Review Experimental units and independence (ai-7/9). Goal: Explain the **mechanism** and terminology of experimental units and independence. Short practice: Count independent plants and photo views.

Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.. Return to this week and explain what you changed.

Review A literature comparison matrix (ai-7/6). Goal: Explain the **mechanism** and terminology of a literature comparison matrix.

Short practice: Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the literature matrix preserve?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

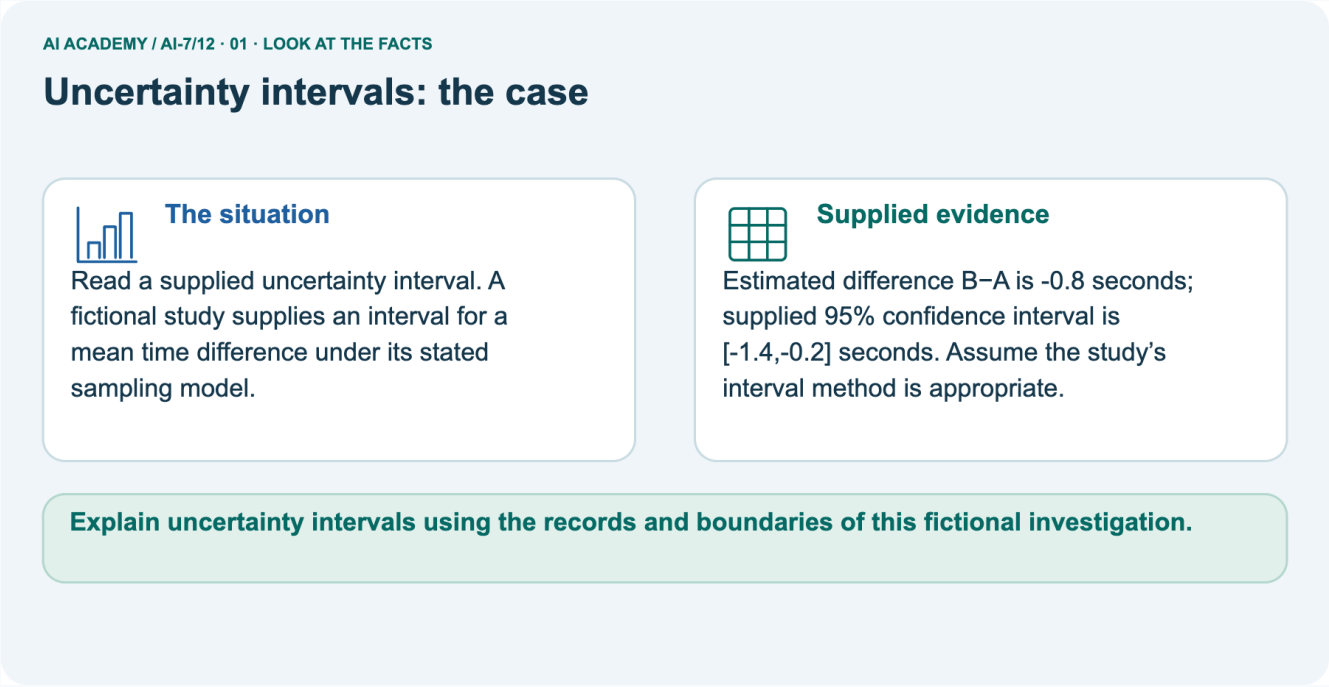
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Uncertainty intervals picture from its labels, then record a relevant limit.

01. Uncertainty intervals: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

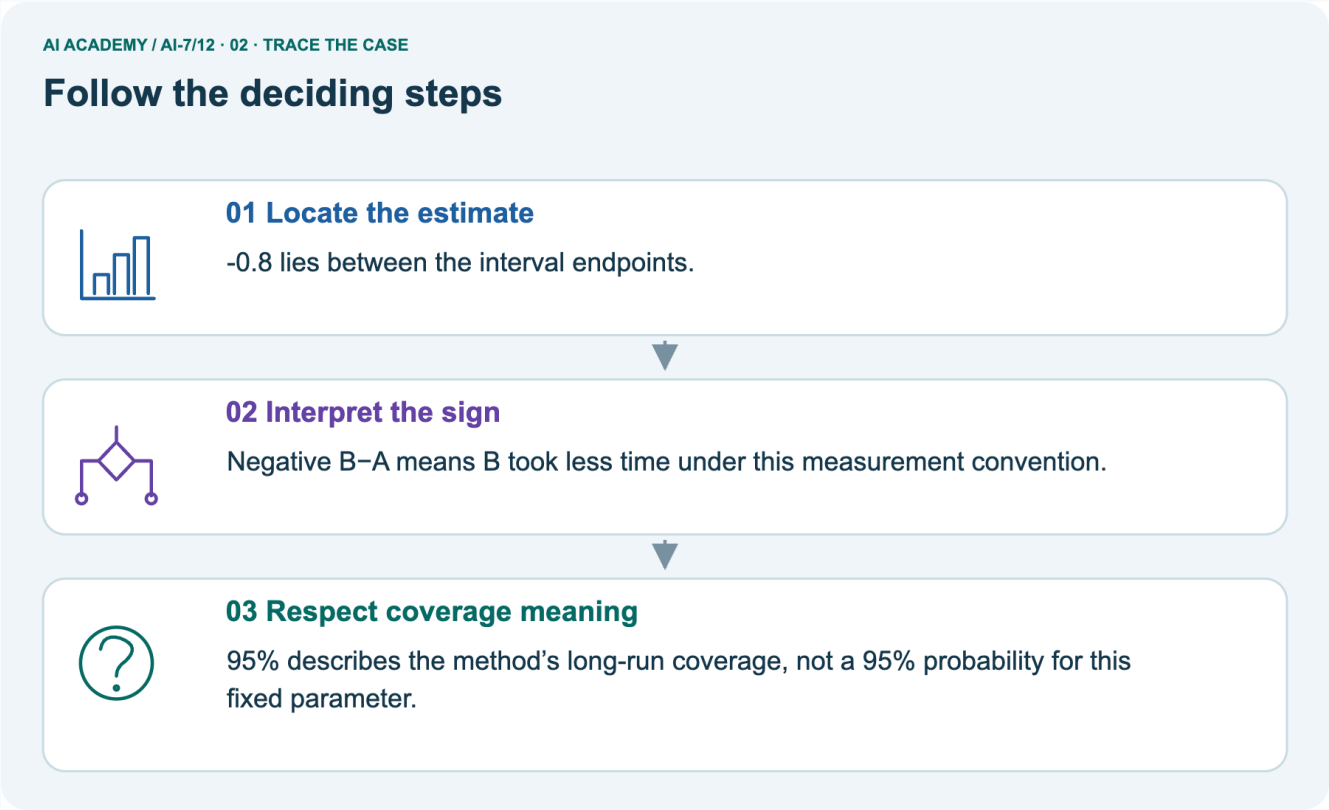
Read a supplied uncertainty interval. A fictional study supplies an interval for a mean time difference under its stated sampling model.

Supplied facts: Estimated difference B−A is -0.8 seconds; supplied 95% **confidence interval** is [-1.4,-0.2] seconds. Assume the study's interval method is appropriate.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



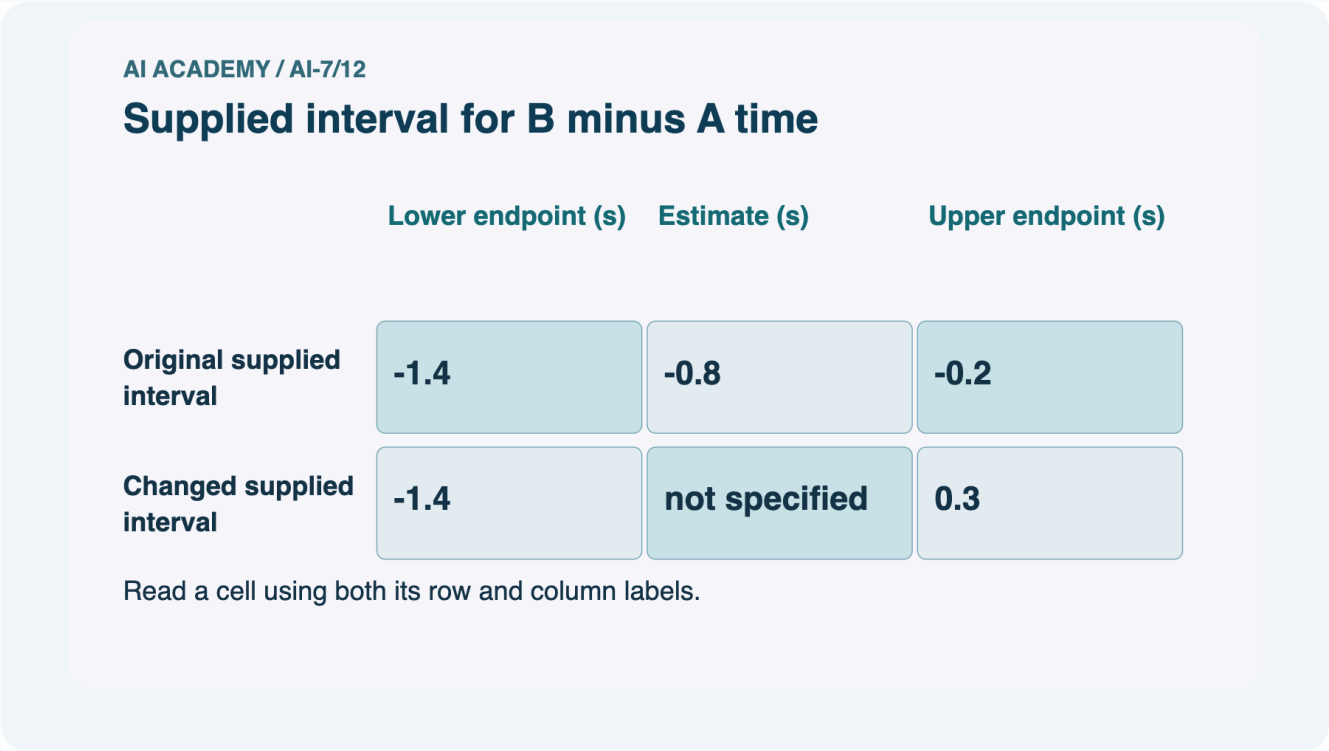
Follow the labelled picture with the matching explanation. Open the vector diagram

Locate the estimate: -0.8 lies between the interval endpoints. → Interpret the sign: Negative B–A means B took less time under this measurement convention. → Respect coverage meaning: 95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Supplied interval for B minus A time



Follow the labelled picture with the matching explanation. Open the vector diagram

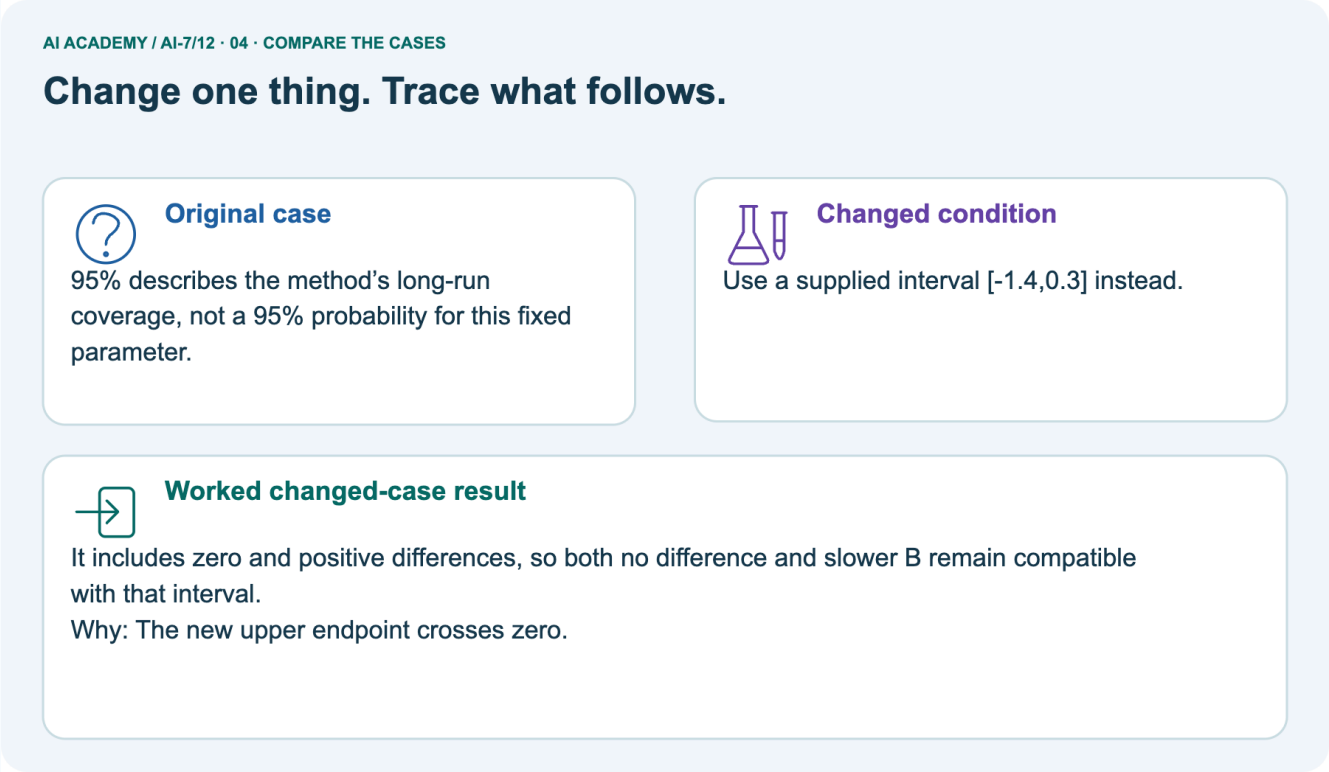
Supplied interval for B minus A time. Original supplied interval: Lower endpoint (s) = -1.4, Estimate (s) = -0.8, Upper endpoint (s) = -0.2.

Changed supplied interval: Lower endpoint (s) = -1.4, Estimate (s) = not specified, Upper endpoint (s) = 0.3.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

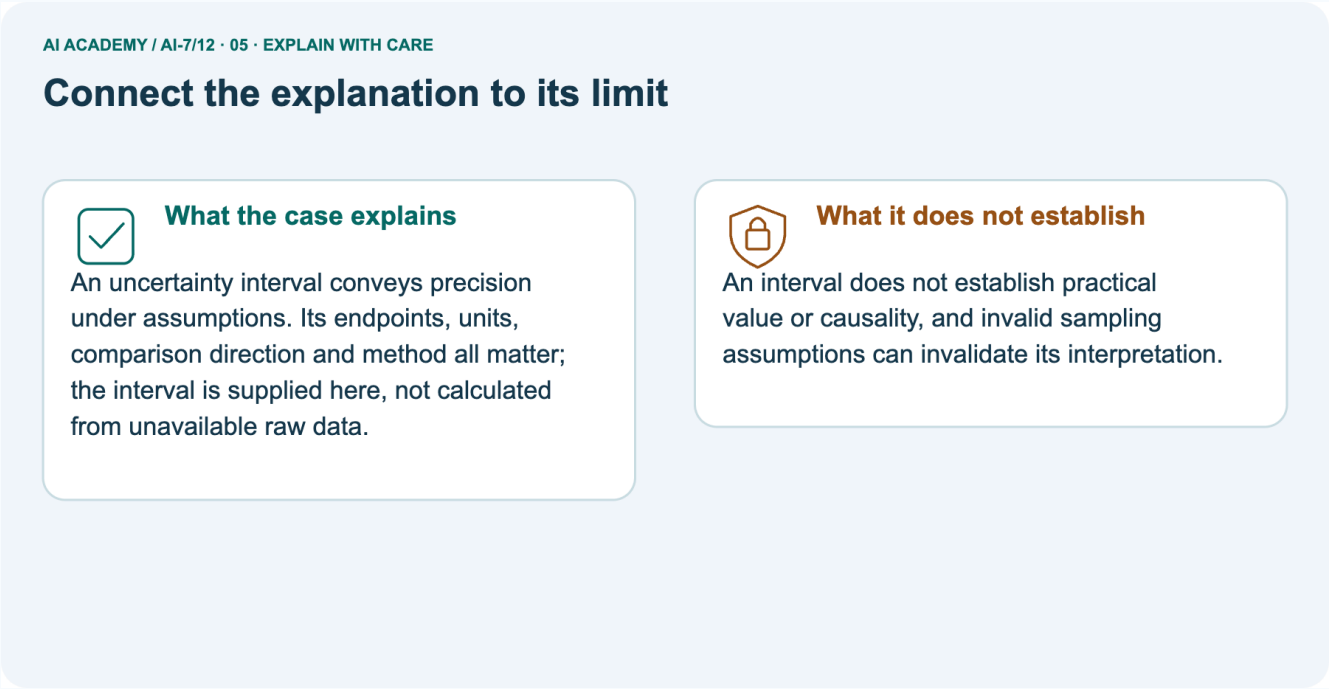
Original case: 95% describes the method's long-run coverage, not a 95% probability for this fixed parameter. Changed condition: Use a supplied interval [-1.4,0.3] instead.

Expected result: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval. Reason: The new upper endpoint crosses zero.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

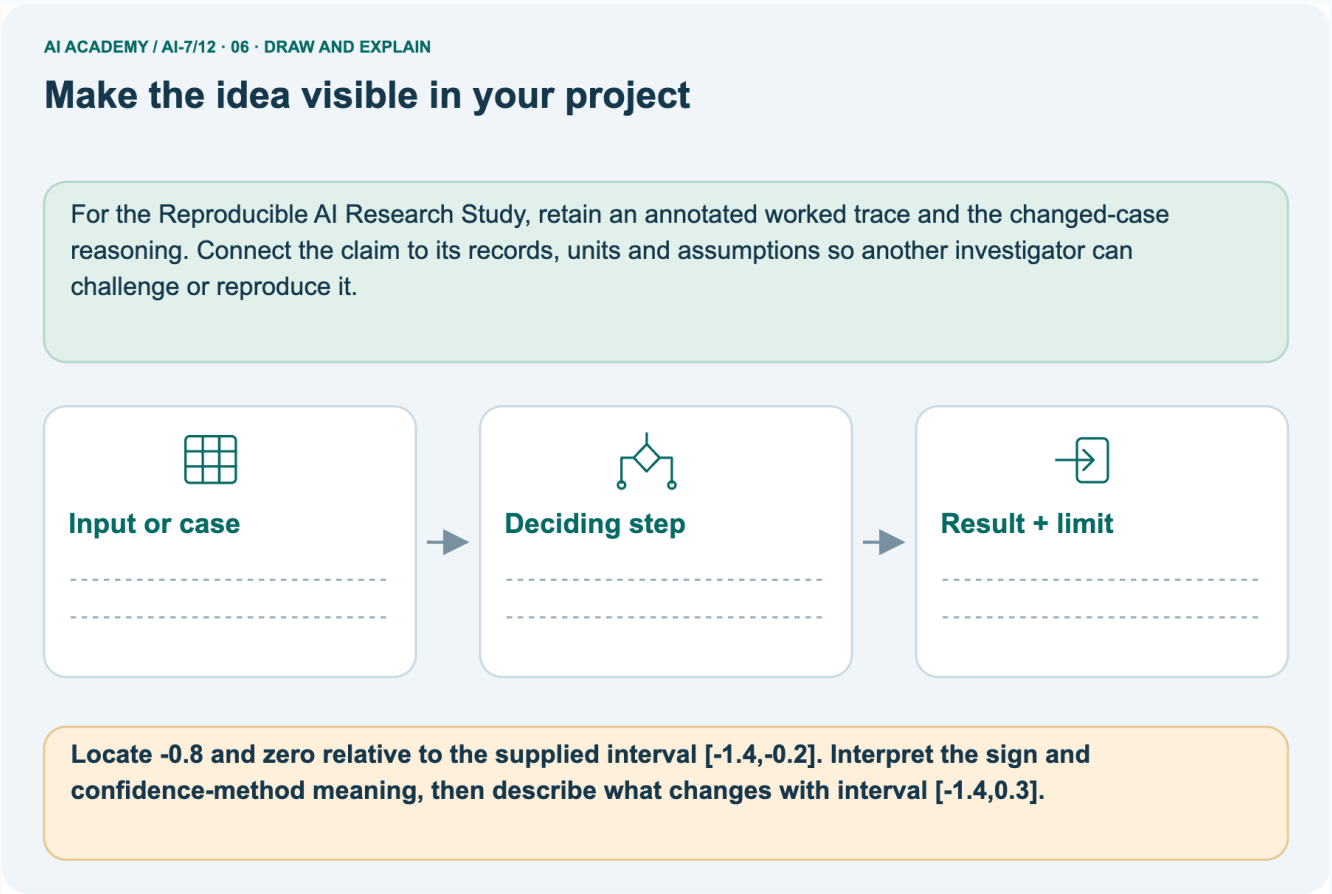
An uncertainty interval conveys precision under assumptions. Its endpoints, units, comparison direction and method all matter; the interval is supplied here, not calculated from unavailable raw data.

Limit: An interval does not establish practical value or causality, and invalid sampling assumptions can invalidate its interpretation.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2].

Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.